

A RECONSTRUCTION PROGRAMME IN STRUCTURAL MATHEMATICS

UNIFIED MECHANICS

Final Edition

Coherent Master 2026.7 · Closing Addendum 2026.8 · Finalisation 2026.12

φ

$$(R_{n+1}, C_{n+1}) = (R_n + C_n, R_n) \implies \varphi^2 = \varphi + 1, \quad r = \frac{1}{2\varphi}$$

JOSEPH SHIELDS

ONE INVARIANT · MANY SECTORS · ONE CONSISTENCY TEST



ABOUT THIS EDITION. This is the final coherent edition of the Unified Mechanics programme, rewritten in dependency order rather than chronological order, following the governing 2026.12 finalisation mandate: every result appears after its hypotheses, every physical name appears after its neutral mathematical structure, and every historical conjecture is either proved conditionally, converted into a representation condition, refuted, or removed from the governing theory. It integrates the 2026.7 Coherent Master core, the 2026.8 Closing Addendum on rigidity and identifiability, and the one-axiom, three-space, internal-age, metallic-ratio, and realization-tree results of the late-2026 sessions. The full proofs, machine verifications, and development record remain in the archival master edition; where formulations differ, this edition governs.

STATUS VOCABULARY. Every claim carries exactly one grade, stated once in its heading and once in the final ledger: *Axiom* (model-defining relation), *Proved* (theorem with complete proof from stated hypotheses), *Conditional* (proved under explicitly stated hypotheses not yet established within the framework), *Proposed* (physical identification consistent with the framework but not derived from it), and *No-go* (a proved limitation retained to prevent overclaiming).

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Author's Note

This monograph begins from one operational question: *what must a system contain in order to retain, update, and physically realize its own complete description?*

The answer is developed in layers. A closed whole-shift relation produces the Fibonacci self-coding encoder and its golden fixed ray. A quadratic realization turns internal closure into an action and a binary distinction into three irreducible symmetric components. Those components generate the minimal local refinement lattice; when all three survive faithfully and without excess, its growth dimension is three, while the irreversible execution order supplies time. The later geometry, field sectors, cosmology, and compact closure are representations of this same architecture.

The work distinguishes exact mathematics from physical interpretation. Quotients, fixed rays, quadratic decomposition, execution categories, refinement growth, cocycle invariants, routing identities, and lattice spectra are graded by proof. Light, matter, gravity, black holes, cosmological read-out, and metallic closure branches are named only after the neutral structures they are proposed to realize.

The final question is therefore not whether an elegant formula resembles Nature, but whether Nature instantiates the finite representation conditions stated by the theory.

Statement of the Programme

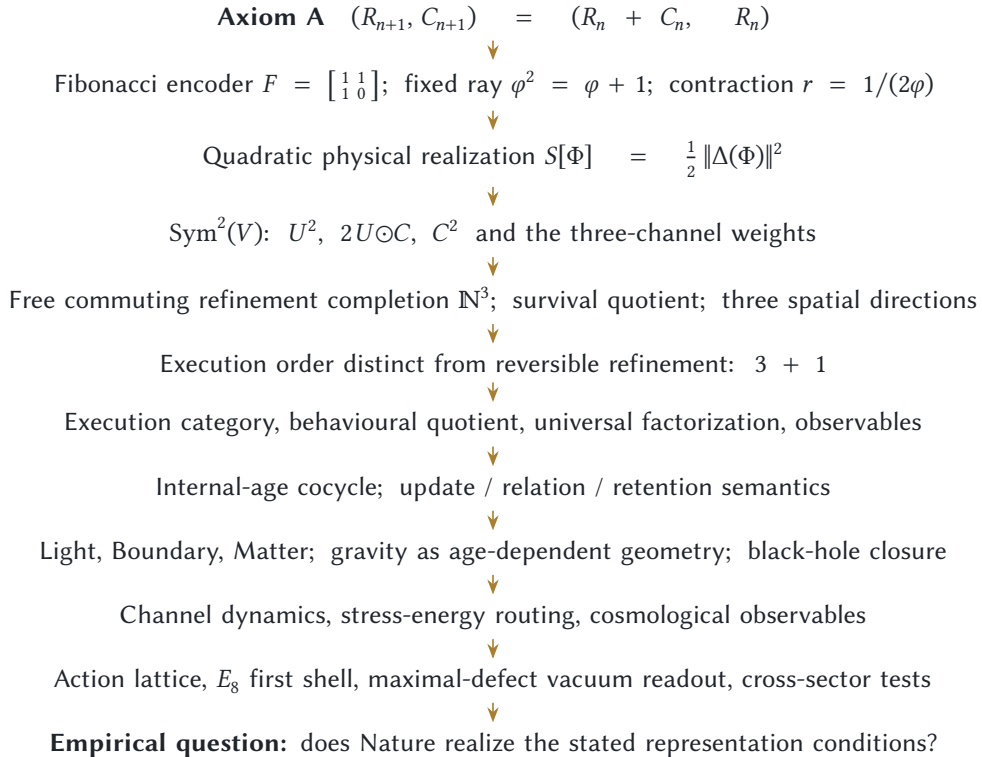
Unified Mechanics is a reconstruction programme whose objective is to determine whether self-description, physical action, spatial dimension, geometry, observables, and the principal sectors of physics can be reconstructed from one internally closed operational relation and its faithful physical realizations.

In one sentence: *Unified Mechanics is a reconstruction programme in which a system that internally retains and updates its own complete description is represented physically through a quadratic closure, from which invariant quotients, three surviving spatial refinements, execution dynamics, internal age, field sectors, and testable cosmological relations are constructed.*

The golden ratio is not the thesis. It is the unique positive fixed ray of the canonical whole-shift realization, derived within Axiom A rather than asserted as a consequence of generic self-description. The thesis is the reconstruction mechanism.

The final dependency chain

Every later chapter is organized by the chain below; each layer appears only after its hypotheses.



THE COMMON HINGE

The quadratic is the bridge between the internally closed axiom and physical observation. It simultaneously turns closure defect into an action and turns a binary internal distinction into three irreducible symmetric components. This is why the one-axiom result and the three-

space result belong in the same early chapter sequence rather than in remote appendices.

PART I

Internal Closure



The Closed Whole-Shift Law

AXIOM A Closed whole-shift self-measurement

[Axiom / model-defining relation]

The internally coded system updates by

$$(R_{n+1}, C_{n+1}) = (R_n + C_n, R_n).$$

The next referent is the previous whole; the next code is the previous referent. This is the canonical internally closed relation used by the theory. Generic self-description *without* this relation does not force the golden law.

THEOREM 1.1 Golden fixed ray

[Proved]

The whole-shift matrix $F = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ has the unique positive projective eigenvalue

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \varphi^2 = \varphi + 1,$$

with normalization and contraction coordinate

$$1 = \frac{1}{\varphi} + \frac{1}{\varphi^2}, \quad r = \frac{1}{2\varphi}, \quad 0 < r < \frac{1}{2}.$$

Proof. The characteristic polynomial of F is $\lambda^2 - \lambda - 1$, with roots φ and $-1/\varphi$; only φ is positive, and its eigenray is the unique attracting projective fixed point. The normalization identity is $\varphi^2 = \varphi + 1$ divided by φ^2 . ♦

NO-GO 1.2 Rewriting alone does not determine the encoder

[Proved]

The general execution environment — a rewrite system satisfying the axioms R1–R4 (termination, local confluence, and their companions) — contains no encoder operation. R1–R4 alone therefore cannot determine the whole-shift law, the value of φ , or the contraction r . The golden structure enters only through Axiom A and its faithful realization.

The rewrite axioms are nevertheless indispensable downstream: under termination and local confluence, Newman's lemma gives a unique normal form $\kappa(X)$ for every presentation X , and the quotient by $\kappa(X) = \kappa(Y)$ removes distinctions the system cannot preserve under complete reduction. Part IV develops that machinery as the invariant completion of the primitive realization constructed next.

PART II

Physical Realization

The Quadratic Realization

Let ρ be a physical representation of the internally closed system and let $\Delta(\Phi)$ be its *closure defect*: the failure of a configuration Φ to satisfy internal closure under ρ .

THEOREM 2.1 Quadratic realization

[Proved under stated functional hypotheses]

For closure defect $\Delta(\Phi)$, the nonnegative physical action

$$S[\Phi] = \frac{1}{2} \|\Delta(\Phi)\|^2$$

has Euler equation

$$D\Delta[\Phi]^* \Delta(\Phi) = 0$$

together with the total (summed) boundary functional. Under the stated nondegeneracy condition, stationary configurations are exactly closure configurations.

NO-GO 2.2 Residuals without nondegeneracy

[Proved]

Without nondegeneracy, the normal equation $D\Delta^* \Delta = 0$ can possess nonzero residuals: stationarity does not then imply closure. This qualification is retained; it marks exactly where the variational reading of execution stops being automatic.

This normal-equation construction is the entry point of the Principle of Equivalent Action: the broader Helmholtz–Vainberg analysis classifies which represented operator classes admit an action whose Euler–Lagrange equations reproduce the dynamics. Variationality is proved for the represented self-adjoint / quadratic operator class; it is not universalized to every rewrite system.

PART III

Triadic Algebra and Three-Space

The Quadratic Triad

Let V be a binary update–retention module with basis $\{U, C\}$ and normalized weights $1 - r$ and r .

THEOREM 3.1 Quadratic triad and boundary polarization

[Proved]

The exchange-blind quadratic closure $\text{Sym}^2(V)$ has exactly three basis types, U^2 , $U \odot C$, and C^2 , with weights

$$(1 - r)^2 + 2r(1 - r) + r^2 = 1.$$

The mixed component is uniquely recovered by polarization and is therefore the *relation* completing the two pure terms — generated by the quadratic, not independently inserted.

Note the distinction the final edition maintains throughout: $\text{Sym}^2(V)$ has dimension three, while the normalized weight simplex is two-dimensional. Three probabilities do not, by themselves, imply three-space. Three-space requires the survival theorem below.

The three exact execution weights, in channel notation,

$$W_L = (1 - r)^2, \quad W_B = 2r(1 - r), \quad W_M = r^2,$$

receive their physical names only in Part V, after the neutral structures they are proposed to realize.

Emergent Space and Time

Define a *spatial direction* as an independent, local, commuting, unbounded refinement generator with positive stable translation length.

THEOREM 4.1 Three-dimensional survival

[Conditional under explicit realization hypotheses]

Let the three primitive quadratic components be represented as distinct local commuting refinement generators. If the realization is faithful, exhaustive, and scale-faithful, then the surviving refinement monoid is $\mathbb{N}^3$ and its group completion is $\mathbb{Z}^3$. The refinement ball satisfies

$$|B_R| = \binom{R+3}{3} \sim \frac{R^3}{6},$$

so the polynomial growth degree and the local continuum dimension are three, and the rescaled group completion tends to $\mathbb{R}^3$.

Time is not a fourth spatial generator. It is the irreversible, well-founded order of *execution* — distinct in kind from reversible commuting translation. The signature of the reconstruction is therefore $3 + 1$ by construction: three surviving refinements plus one execution order.

PART IV

Execution and Invariance

Execution Category and Quotients

With the primitive local generators supplied by Parts I–III, the execution machinery removes representational redundancy.

Under termination and local confluence, every presentation X has a unique normal form $\kappa(X)$, and $X \sim Y \iff \kappa(X) = \kappa(Y)$ defines the behavioural quotient $Q(E)$ of the execution category E . The quotient carries a universal factorization property: every observable constant along reduction factors uniquely through $Q(E)$, and equivariant group actions descend. Topology, metric structure, and the observable algebra are then constructed on the quotient, not on presentations.

The categorical results are exact and unconditional; their role in the final edition is precisely stated: *they are the invariant completion of the primitive quadratic realization*, removing redundancy after the local generators exist — they do not, by themselves, generate the encoder, the triad, or three-space.

PART V

Internal Age and Field Semantics

Internal Age

THEOREM 6.1 Internal-age well-definedness

[Proved]

An additive age increment with zero circulation on every confluence cell defines a path-independent internal age τ on the execution quotient. Age is retained execution history, not external clock time.

The cocycle condition is the whole content: path independence on the quotient is exactly zero circulation on confluence cells, and τ descends because behaviourally equivalent presentations reduce through joinable paths. Two further neutral notions are fixed here, before any physical naming:

- *age-neutral propagation* — generators along which τ does not accumulate, with positive stable translation length;
- *persistent retained history* — states whose accumulated τ is positive and stable under further execution.

Execution time (the well-founded order of Part III) and internal age (the retained-history cocycle) are distinct and are never merged.

Light, Boundary, Matter, and Gravity

The neutral definitions come first; the physical names are realization conditions.

THEOREM 7.1 Physical semantic realization

[Conditional / physical identification]

Under the physical representation: age-neutral generators with positive stable length are light-like propagation; persistent positive age is matter-like retention; the polarized mixed sector is the boundary relation; and age-dependent metric cost is gravitational geometry. The physical names remain representation conditions, while the age cocycle, kernel, and persistence statements are mathematical.

Term	Final meaning
Update	A local operation that changes relational or topological state.
Retention / contraction	The part of execution history encoded persistently in the current state.
Boundary	The polarized cross-relation required for update and retention to close as a quadratic whole; also the summed interface condition.
Internal age τ	A quotient-invariant measure of retained contraction history.
Light	The physical realization of age-neutral propagation with positive stable translation.
Matter	The physical realization of persistent retained update-history.
Gravity	The physical realization of metric or potential response to internal-age variation.
Black hole	A proposed age-saturated state whose outward update capacity vanishes relative to the parent realization.
Child universe	A proposed new realization whose initial whole is the image of a terminal parent closure.
Heaviest stone	A proposed extremal realization maximizing stable contraction depth while preserving coherent closure.

Table 7.1: The final semantic dictionary. Names in the lower rows are graded in Parts VI and X.

The compressed chain that the remainder of the monograph unfolds:

update $\rightarrow$ retained update $\rightarrow$ internal age $\rightarrow$ age-dependent geometry $\rightarrow$ terminal closure $\rightarrow$ new

realization

PART VI

Terminal Closure and Realization Trees

Black Holes and Realization Trees

Define the *outward update capacity* of a state as the stable translation available from it toward its parent realization, and *age saturation* as the exhaustion of further retainable contraction.

THEOREM 8.1 Black-hole closure system

[Conditional theorem and physical proposal]

A black-hole state is represented by age saturation together with vanishing outward stable update capacity. A parent–child realization exists when the terminal parent closure is mapped to an initial child whole by a closure functor,

$$B_{\text{terminal}} \mapsto I_{\text{child}}.$$

This is a precise conditional architecture. The claim that astrophysical black holes realize it is empirical.

Ordinary container language is avoided deliberately: a child is not asserted to sit spatially inside its parent; a terminal parent normal form is mapped to an initial child whole. The *heaviest-stone principle* — that the realized branch maximizes stable retained depth compatible with continued closure — is an extremal conjectural semantics. It is not required by the formal core and is graded accordingly in the final ledger.

PART VII

Cosmology and Measurement

Channel Dynamics and Conservation

THEOREM 9.1 Channel dynamics and whole conservation [Proved in the tensor execution model]

Tensor composition gives r^k mobility on the k -defect sector; additive one-body defect counting gives a km_*^2 gap. Interactions may exchange channel content, but the total relation obeys

$$\sum_k J_k = 0.$$

The boundary condition and the conservation law belong to the whole sum, not necessarily to each component separately.

Complete stress tensors — not only scalar weights — are routed through the column-stochastic map of the next chapter; the exchange currents J_k are how sectors trade content while the whole remains conserved.

Cosmological Routing

THEOREM 10.1 Cosmological routing

[Conditional under explicit readout and calibration]

The relational registration machine has the unique routing matrix

$$T = \begin{pmatrix} 1 & 1-2r & \frac{1}{2} \\ 0 & 2r & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

With common source calibration it yields

$$\Omega_b = \frac{r^2}{2}, \quad \Omega_{\text{DM}} = 4r^2(1-r), \quad \Omega_{\text{DE}} = 1 - \frac{9}{2}r^2 + 4r^3,$$

and equal calibration is *necessary and sufficient* for the displayed fractions.

The densities are therefore not arbitrary r -polynomials chosen to fit: they are the unique routed outputs of one relational machine under one energy calibration. What remains physical input is exactly that machine and that calibration.

NO-GO 10.2 A fixed background does not determine perturbations

[Proved]

Let $P_0(X, \phi)$ admit a background trajectory $X = X_0(\phi)$. For any sufficiently regular $\alpha(\phi)$, the deformation

$$P_\alpha(X, \phi) = P_0(X, \phi) + \alpha(\phi)(X - X_0(\phi))^2$$

retains the same background solution, yet $(P_\alpha)_{XX}|_{X_0} = (P_0)_{XX}|_{X_0} + 2\alpha(\phi)$, so the scalar sound speed

$$c_s^2 = \frac{P_X}{P_X + 2XP_{XX}}$$

generally changes. A background equation of state does not uniquely determine perturbation propagation. The framework presents a stable canonical scalar representative while retaining this non-uniqueness.

10.1 The three-direction braid and the Hubble question

THEOREM 10.3 Three-direction braid scalar

[Proved scalar identity; physical projection open]

For isotropic primitive contraction $B_3 = rI_3$, the unique linear basis-invariant scalar is

$$\text{Tr}(B_3) = 3r.$$

This does not by itself equal an observed Hubble discrepancy. A physical projection

$$\Delta_H = \Pi_H(B_3, \tau_{\text{observer}}, \tau_{\text{source}})$$

from the braid and the observer/source internal ages to the inferred expansion shift must be derived or measured.

The final edition may propose that local and background expansion estimates are different projections of one update-age braid; it does not identify the observed discrepancy directly with $3r$ without an observation theorem. Deriving or falsifying Π_H is a named research target, not a claim.

Quantum Registration and the Action Lattice

THEOREM 11.1 Quantum registration and action

[Proved / conditionally calibrated]

The normalized Perron frequencies of the golden registration are

$$\left(\frac{1}{\varphi}, \frac{1}{\varphi^2} \right)$$

and define an ordinary Born-compatible two-outcome instrument. No automatic Born-rule deviation remains. Three independent completion coordinates define a relative coherence cell with dimensionless volume r^3 , and one physical calibration

$$S_* r^3 = \hbar$$

fixes the dimensionful action unit.

Two superseded formulations are permanently replaced here: golden routing is a Born-compatible POVM (no deviation follows without a separate comparator model), and quantization proceeds through the relative three-channel coherence cell — termination forbids the directed execution cycles of the earlier construction.

PART VIII

Internal Carrier and Vacuum

E_8 and the Vacuum Hierarchy

THEOREM 12.1 Internal carrier and vacuum hierarchy

[Conditional under carrier and readout

conditions]

Three binary completion coordinates require an eight-dimensional minimal faithful outcome carrier. A closed spin positive even unimodular rank-eight pairing is E_8 . The first nonzero lattice shell contains 240 root modes. For a monoidal frozen root state and the unique maximal-defect projector

$$\Pi_{\max} = P_b^{\otimes 240},$$

the joint expectation is r^{240} . Its interpretation as ρ_Λ / M_*^4 is a physical vacuum-readout condition, not a generic determinant result.

Rank eight is derived from the faithful three-coordinate outcome algebra, not assumed by minimality; the exponent is the internal first-shell count 240 (a 241st factor would require an explicitly external readout, and is excluded from the governing theory); and the Gaussian-determinant shortcut to vacuum energy is deleted — a determinant enters only through $\log \det$, and the governing construction is the maximal-defect expectation / selection weight.

PART IX

Metallic Closure Branches

Metallic Ratios and the Silver Branch

THEOREM 13.1 Metallic closure-arity spectrum

[Spectral theorem; physical branch conditional]

For the n -return closure matrix $F_n = \begin{bmatrix} n & 1 \\ 1 & 0 \end{bmatrix}$, the positive fixed ratio is

$$\mu_n = \frac{n + \sqrt{n^2 + 4}}{2}, \quad \mu_n^2 = n \mu_n + 1.$$

$n = 1$ gives the golden self-description branch. $n = 2$ gives the silver two-return branch,

$$F_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad \delta = 1 + \sqrt{2}, \quad \delta^2 = 2\delta + 1.$$

Metallic ratios are a closure-arity family — eigenvalues of specific return matrices, not decorative constants. The astrophysical hypothesis is phrased as a theorem target: galactic rotation realizes the silver branch only if two independent return relations are constructed in the galactic execution quotient *and* the circular-velocity observable is derived from the Perron mode. Until both are supplied, the identification is a conditional realization programme.

PART X

Audit, Falsification, and the Final Ledger

Cross-Sector Rigidity and Identifiability

A framework is not unified merely because the same symbol appears in several formulas. The same invariant must be recoverable independently from several sectors, and those recoveries must agree; otherwise the symbol silently becomes a different fitted parameter in every application. With

$$p = 2r, \quad h = 1 - 2r = p^2, \quad b = \frac{r^2}{2}, \quad d = 4r^2(1 - r), \quad e = 1 - b - d, \quad q = r^3, \quad \lambda = r^{240},$$

the question is: how many independent numbers are these? The answer is one.

THEOREM 14.1 Cross-sector closure identities

[Proved within the routed UM model]

On the golden branch,

$$h = 8b, \quad q^2 = 8b^3, \quad \lambda = q^{80} = (2b)^{120},$$

$$d = 8b(1 - \sqrt{2b}), \quad e = 1 - b - d,$$

and, eliminating the square root,

$$(8b - d)^2 = 128b^3, \quad 0 < d < 8b.$$

Proof. Since $p = 2r$ and $h = p^2$, $h = 4r^2 = 8b$. Since $q = r^3$ and $b = r^2/2$, $q^2 = r^6 = (2b)^3 = 8b^3$; likewise $\lambda = r^{240} = (r^3)^{80} = q^{80} = (r^2)^{120} = (2b)^{120}$. For dark matter, $r = \sqrt{2b}$ on the physical branch, so $d = 8b(1 - \sqrt{2b})$ and $8b - d = 8b\sqrt{2b}$; squaring gives $(8b - d)^2 = 128b^3$, with $0 < d < 8b$ selecting the physical root. ♦

COROLLARY 14.2 Registration alone fixes the background dictionary

[Proved]

If p is measured and $p + p^2 = 1$ holds, then

$$r = \frac{p}{2}, \quad b = \frac{p^2}{8}, \quad h = p^2, \quad q = \frac{p^3}{8}, \quad \lambda = \left(\frac{p}{2}\right)^{240}, \quad d = p^2\left(1 - \frac{p}{2}\right), \quad e = 1 - b - d.$$

COROLLARY 14.3 Baryonic fraction fixes the hierarchy

[Proved]

On the physical branch $r = \sqrt{2b}$, hence

$$\frac{\rho_\Lambda}{M_\star^4} = (2\Omega_b)^{120}$$

under the vacuum identification: the baryonic routing fraction and the vacuum hierarchy are not independent assignments. Likewise the channel simplex is fixed by b alone:

$$W_M = 2b, \quad W_B = 2\sqrt{2b}(1 - \sqrt{2b}), \quad W_L = (1 - \sqrt{2b})^2.$$

14.1 Independent estimators of one invariant

Define the sector estimators

$$\hat{r}_{\text{reg}} = \frac{\hat{p}}{2}, \quad \hat{r}_b = \sqrt{2\hat{b}}, \quad \hat{r}_q = \hat{q}^{1/3}, \quad \hat{r}_\Lambda = \hat{\lambda}^{1/240},$$

and let $\hat{r}_d$ be the unique solution in $(0, \frac{1}{2})$ of $4r^2(1 - r) = \hat{d}$.

THEOREM 14.4 Uniqueness of the dark-matter estimator

[Proved]

$f(r) = 4r^2(1 - r)$ is strictly increasing on $(0, \frac{1}{2})$, since $f'(r) = 4r(2 - 3r) > 0$ there; every d in the image determines a unique r_d .

THEOREM 14.5 Diagonal consistency criterion

[Proved]

The cross-sector realization is consistent exactly when the admissible uncertainty regions for $\hat{\mathbf{r}} = (\hat{r}_{\text{reg}}, \hat{r}_b, \hat{r}_d, \hat{r}_q, \hat{r}_\Lambda)$ possess a common intersection; in the zero-uncertainty idealization,

$$\hat{r}_{\text{reg}} = \hat{r}_b = \hat{r}_d = \hat{r}_q = \hat{r}_\Lambda.$$

Allowing a separate value in each sector replaces one theory by five unrelated one-parameter descriptions.

PROPOSITION 14.6 Codimension of unification

[Proved]

If n sectors are each allowed an independent effective contraction r_i , their unconstrained parameter space is $(0, \frac{1}{2})^n$; the unified condition $r_1 = \dots = r_n$ is the diagonal, of codimension $n - 1$. Every additional readable sector adds a genuine constraint, not a fitting freedom.

For empirical work, with propagated uncertainties σ_i and the weighted common estimator $\bar{r} = (\sum_i \hat{r}_i / \sigma_i^2) / (\sum_i 1 / \sigma_i^2)$, the statistic

$$\chi_{\text{UM}}^2 = \sum_i \left(\frac{\hat{r}_i - \bar{r}}{\sigma_i} \right)^2$$

is the direct test of the single-invariant claim once measurements and covariances are supplied.

14.2 Parameter-free is not representation-free

THEOREM 14.7 Representation-moduli criterion for derivation

[Proved]

Let $\mathfrak{R}$ be the class of representations satisfying the stated anchors, modulo physically irrelevant equivalence, and $\mathcal{O} : \mathfrak{R} \rightarrow Y$ a predicted observable. $\mathcal{O}$ is rigidly derived only if it is constant on $\mathfrak{R}$; two admissible representations with $\mathcal{O}(R_1) \neq \mathcal{O}(R_2)$ are a countermodel to uniqueness.

COROLLARY 14.8 Honest zero-parameter criterion

[Proved]

A sector is zero-free-parameter in the strong physical sense only when its admissible representation moduli leave all claimed dimensionless observables invariant. Discrete structural anchors are not fitted numbers, but unclassified functional freedom inside an anchor cannot be ignored.

THEOREM 14.9 No rescue by silent refitting

[Proved]

If an exact elimination identity fails and the failure is repaired only by assigning a distinct contraction parameter to that sector, the repaired model is not the same unified theory: it is an extension with at least one additional independent parameter, by Proposition 14.6, and must be graded as such.

PROPOSITION 14.10 Vacuum hierarchy sensitivity

[Proved]

For $\lambda = r^{240}$, $d \log \lambda / d \log r = 240$, so $\delta \lambda / \lambda \approx 240 \delta r / r$: the hierarchy is an exceptionally sharp cross-sector test once the same normalization and physical vacuum observable are fixed.

Supersession and Deletion Ledger

The following formulations are permanently superseded in the governing theory. Older statements survive only in the development record of the archival master edition.

Old or risky formulation	Action	Final governing form
Generic self-description forces φ	Replace	Only the closed whole-shift relation forces the Fibonacci encoder; the no-go for generic self-description is preserved.
φ is an independent algebraic axiom	Reframe	φ is derived from Axiom A in the one-axiom narrative; the earlier separate-branch formulation is historical.
Three channels imply three-space	Replace	The survival theorem governs: faithful, exhaustive, unbounded commuting refinement generators yield $\mathbb{N}^3$ and growth degree three.
Boundary is a third substance	Delete	Boundary is the polarized mixed relation and total interface condition; substrate language is secondary.
All execution dynamics is variational	Restrict	Variationality is proved for the represented self-adjoint / quadratic operator class only.
Every rewrite refinement becomes smooth Lorentzian spacetime	Delete	Explicit model classes and a defined local covariant realization condition are used instead.
Cereal-bowl 70/25/5 rule	Remove	Retained only as historical intuition unless a basin-trap theorem is supplied.
Born-rule deviation of 13.6 ppm	Delete	Golden routing is Born-compatible; no deviation follows without a separate comparator model.
Directed execution-cycle quantization	Delete	Refuted: termination forbids directed cycles; the relative three-cell coherence construction governs.
Hubble shift = $3r^3$	Delete	Three-output additivity does not define an observation shift.
Hubble shift = $3r$ directly	Restrict	$3r$ is a trace invariant; the observation map Π_H is a physical theorem target.
Unconditional $c = \varphi$ braiding	Delete	Retained only in representations identifying the physical braid operator with the Fibonacci encoder.
Vacuum energy from generic determinant ratio r^{240}	Delete	A Gaussian determinant enters through log det; the maximal-defect expectation / selection weight governs.
r^{241} internal E_8 count	Delete	The internal first shell is 240; a 241st factor requires an explicitly external readout.
Cosmological densities as arbitrary r -polynomials	Replace	Derived as the unique routed outputs of the relational machine under common energy calibration.
Silver ratio explains galactic rotation	Grade	The spectral theorem is exact; the two-return galactic realization remains to be constructed and tested.
Black holes create universes	Grade	The parent–child closure functor is precise; astrophysical realization remains empirical.

Table 15.1: The supersession and deletion ledger. No formulation in the left column is active in this edition.

The Final Status and Falsification Ledger

16.1 Final status of every main claim

Claim	Final status	Falsification / obligation
Whole-shift encoder and φ	Proved from Axiom A	Refute by rejecting Axiom A as the physical self-coding realization.
Quadratic action normal equation	Proved under functional hypotheses	Check nondegeneracy and represented operator class.
Triadic Sym^2 decomposition	Proved	Pure algebraic identity.
Three-space	Conditional theorem	Test faithful, exhaustive, unbounded local realization of all three generators.
Internal age cocycle	Proved when confluence closed	Check path independence / zero circulation.
Light / Matter / gravity semantics	Physical representation	Construct observables distinguishing age-neutral propagation, retained history, and age-dependent geometry.
Black-hole closure tree	Conditional architecture / proposal	Construct the parent-child closure map and compare with black-hole physics.
Channel dynamics	Proved in tensor model	Test whether physical sectors realize the tensor defect operator.
Cosmological routing	Conditional theorem	Test common calibration, routed densities, and exchange currents.
$3r$ Hubble braid	Proved trace identity; observational proposal	Derive or falsify Π_H .
Golden registration	Proved Perron frequency; Born-compatible	Test operational self-coding realization.
r^3 action lattice	Proved dimensionless; calibrated physically	Test coherence cell and action scale.
E_8 first shell	Conditional theorem	Test closed spin rank-eight carrier and shell truncation.
r^{240} vacuum	Conditional theorem	Test monoidal root statistics and maximal-defect stress-energy readout.
Silver galactic branch	Spectral theorem + physical proposal	Construct two-return galactic quotient and rotation observable.
Parent multiverse / heaviest stone	Physical proposal	No claim of proof from the formal core.

Table 16.1: The final status ledger. Each claim appears exactly once; no earlier scorecard overrides this table.

16.2 Empirical membership tests: the r -free identities

The framework can be tested without repeatedly solving for r ; the following identities eliminate it entirely.

Bridge compared	Exact relation	Failure means
Direct / deferred registration	$p + h = 1, h = p^2$	The stationary golden registration model fails.
Registration / baryons	$h = 8b$	Registration and routed matter do not share one r .
Baryons / dark matter	$(8b - d)^2 = 128b^3, 0 < d < 8b$	The conserved routing map fails.
Background closure	$b + d + e = 1$	The three-output accounting is incomplete or misidentified.
Baryons / action lattice	$q^2 = 8b^3$	The r^3 coherence unit is not the same contraction as matter routing.
Action / vacuum	$\lambda = q^{80}$	The 240-root vacuum event is not built from the same three-channel unit.
Baryons / vacuum	$\lambda = (2b)^{120}$	The root-shell hierarchy and cosmological routing are inconsistent.
All readable sectors	$r_{\text{reg}} = r_b = r_d = r_q = r_{\Lambda}$	The framework has split into sector-specific effective parameters.

Table 16.2: Empirical membership tests: every row is an r -free identity whose failure has a named meaning. Different sectors are not independently adjustable without defining a different theory.

16.3 The exact golden benchmark

The exact branch gives $\varphi = 1.618033988749894848 \dots$ and $r = 0.309016994374947424 \dots$

Quantity	Golden value
Direct registration p	0.618033988749894848 ...
Deferred registration h	0.381966011250105152 ...
Baryonic fraction b	0.047745751406263144 ...
Dark-matter fraction d	0.263932022500210304 ...
Residual fraction e	0.688322226093526552 ...
Dimensionless action unit $q = r^3$	0.029508497187473712 ...
Vacuum hierarchy $\lambda = r^{240}$	$3.9424607712196638 \times 10^{-123}$
$\log_{10} \lambda$	-122.404232619350383 ...
Light-channel weight W_L	0.477457514062631440 ...
Boundary-channel weight W_B	0.427050983124842272 ...
Matter-channel weight W_M	0.095491502812526288 ...

Table 16.3: The exact golden benchmark. These numbers are not independent successes; they are one point on the cross-sector consistency manifold. The scientific claim is the manifold, not the length of the table.

Final Conclusion

Unified Mechanics closes as a reconstruction, not as a declaration that every physical identification has already been observed. The formal system is internally complete once its model-defining whole-shift and representation conditions are stated. The golden ratio is the fixed ray of self-coding closure. The quadratic is the hinge by which internal consistency becomes physical action and a binary distinction becomes a triad. Three-space is the growth geometry of the three local refinements that survive that quadratic faithfully. Time is the irreversible order in which those spatial states execute.

The physical semantics then compresses to one chain: light is age-neutral update; matter is retained update-history; Boundary is the relation completing them; gravity is the geometry of retained internal age; black-hole closure is the proposed terminal loss of outward updating; and a child realization is the proposed re-reading of terminal closure as an initial whole. Cosmological routing, quantum registration, the action lattice, E_8 closure, and the r^{240} vacuum relation are downstream tests of the same invariant contraction.

The monograph therefore ends with a finite scientific question. Does the universe realize the closed self-coding, local refinement, internal-age, routing, and compact-carrier structures defined here? If it does, the apparently separate sectors are constrained to lie on one algebraic and operational diagonal. If it does not, the failure will occur at a named bridge and will not be hidden by retuning another sector.

In the shortest form: the system closes itself internally; the quadratic makes that closure physical; space is what survives the lattice; light updates; matter remembers; gravity records the age of that memory; and completed states may become new beginnings.

The system closes itself internally.

The quadratic makes that closure physical.

Three-space is what survives the lattice.

Light updates. Matter remembers. Boundary relates. Gravity ages.

Completed states may become new beginnings.



ONE INVARIANT · MANY SECTORS · ONE CONSISTENCY TEST